

Jaden Dew

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Education

University of California, Berkeley

Expected Graduation: May 2026

B.S. Aerospace Engineering, GPA: 3.817/4.0

Relevant Coursework: Intro to Compressible Flows, Combustion Processes, Heat Transfer

Skills

Technical: CFD, CAD, Lathe, Mill, Waterjet, Aerodynamics, Propulsion, Test Operations, Design Optimization

Software: Python, MATLAB, Julia, Siemens NX, Fusion 360, OpenFoam, Ansys, WinPlot, Git, Command Line

General: Communication, Leadership, Teamwork, Organization, Problem Solving

Work Experience

Vast Space | [Propulsion Test Intern](#)

May 2025 – Aug 2025

Mojave, CA

- Conducted vacuum chamber hot-fire test operations of Impulse Space's Saiph thruster resulting in 24+ successful hot-fires, 9 successful test campaigns, and the completion of Haven Demo's flight rationale.
- Managed the flow of hot-fire data and conducted data reviews utilizing LabView and WinPlot diagnosing anomalies and assisting REs with developing fixes resulting in smooth test operation across 9 campaigns.
- Designed and constructed a COPV fill measurement and heating system in Siemens NX integrating it with a high-altitude test stand in support of holistic flight like testing of Haven 1's propulsion system.
- Designed mounting infrastructure for heat exchanger upgrade of a vacuum chamber test stand resulting in the successful installation of the upgrade.

NASA Ames Research Center | [Aerodynamics Research Intern](#)

Jun 2024 - Dec 2024

Mountain View, CA

- Utilized DAFOam to develop 8+ reduced C_d airfoils for Titan's atmosphere across multiple AOAs, Reynolds, and Mach numbers, validating results in OpenFoam over 20+ AOAs at flight representative operating points.
- Developed a 3D shape optimization script in Julia, utilizing it to develop 2 FM enhanced rotor blade geometries resulting in a 2% overall increase in FM during hover confirmed in NASA OVERFLOW CFD.
- Presented results to NASA's Dragonfly team pitching this optimization pipeline for future rotor designs.

FLOW Lab – Dr. Simo Mäkiharju | [Undergraduate Research Assistant](#)

Feb 2024 - May 2024

Berkeley, CA

- Developed surface roughness measurement procedure for superhydrophobic surfaces utilizing a Mitutoyo SJ-210 surface tester resulting in the quantifying of surface characteristics of 15+ surfaces.
- Created a script in MATLAB to filter surface profile measurements to calculate R_a , R_{ku} , R_{sk} , and sand grain roughness utilizing those values to correlate surface roughness to hydrodynamic drag reduction.

Project Experience

Space Enterprise at Berkeley | [Propulsion Engineer](#)

Sep 2022 - Present

- Designed and fabricated the main valve assembly for a vertical propulsion system test-stand resulting in multiple successful cold-flows and 3+ successful hot-fire tests of Eureka-3's propulsion system.
- Own the design of a nitrous side custom poppet main valve for Eureka-3's flight propulsion system including CAD, FEA, and pressure loss sims resulting in multiple successful cold-flows.
- Owned the maintenance, assembly, and integration of the Lightbulb engine for the Eureka-2 liquid rocket including flight and ground testing operations resulting in 3 successful hot-fires and a successful launch.